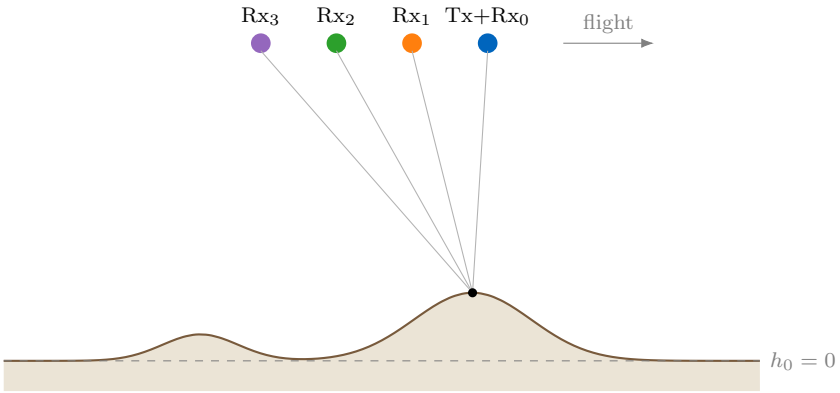
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Topography-Dependent Azimuth Reconstruction in Multistatic SAR Systems

Project report



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Abstract

Multichannel and multistatic synthetic aperture radar (SAR) systems for high-resolution wide-swath (HRWS) imaging sample the azimuth signal of each receiver at a pulse repetition frequency (PRF) below the Doppler bandwidth and recover the unambiguous spectrum by combining the channels in an azimuth reconstruction. The reconstruction filters are derived for a reference geometry. When the receivers are separated by cross-track baselines, a scatterer at a height different from the reference height carries a channel-dependent phase residual, the reconstruction no longer cancels the aliased parts of the spectrum, and azimuth ambiguities appear.

This report studies that topography-dependent mismatch with a numerical simulator of a spaceborne formation ($\lambda = 0.25\,\mathrm{m}$, $r_0 = 766\,\mathrm{km}$, $N_{\mathrm{rx}} = 2$ to 8 receivers, cross-track baselines up to 500 m) and compares correction strategies that use *only the system parameters and a digital elevation model (DEM)*, never the positions of the targets. The sub-aperture topography- and aperture-dependent (SATA) algorithm, applied to each channel before the reconstruction, is shown to be exact only when the terrain has the same height at every ground position folded onto the same Doppler bin, i.e. at positions $X \approx 6.2\,\mathrm{km}$ apart; on general terrain it leaves a structural residual that is predicted from the geometry within about 1 dB. A *joint* reconstruction that places the DEM phase of every folded parcel inside the inversion removes this limitation. Its final version models the channels from the focused line and the DEM and solves a regularised least-squares problem with conjugate gradients.

Over 1040 one-dimensional runs (52 scenes, $N_{\mathrm{rx}} = 2$–8) the median highest ambiguity drops from $-8.4\,\mathrm{dB}$ without correction to $-40.6\,\mathrm{dB}$ with the joint reconstruction (reference $-47.6\,\mathrm{dB}$). The two-dimensional extension, which applies the DEM at every pixel of a 3-D scene, reaches a median of $-44.4\,\mathrm{dB}$ over 640 runs (no correction $-13.9\,\mathrm{dB}$, reference $-47.6\,\mathrm{dB}$). The remaining failures are configurations whose baselines do not separate the folded parcels (two channels, equally spaced baselines) and are a property of the data rather than of the algorithm.

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Symbols and acronyms

Symbol	Meaning
λ	wavelength (0.25 m)
v	platform velocity
r_0	slant range of closest approach of the reference point
θ	look angle ($\approx 20°$)
N_{rx}	number of receive channels (one transmitter, N_{rx} receivers)
$b_{\mathrm{at},i}$, $b_{\mathrm{xt},i}$	along-track and cross-track baseline of receiver i relative to the transmitter
PRF	full (reconstructed) pulse repetition frequency, 2 000 Hz
$\mathrm{PRF}_{\mathrm{op}}$	PRF of each channel, $\mathrm{PRF}/N_{\mathrm{rx}}$
B_{a}	processed azimuth (Doppler) bandwidth, 1 235 Hz
$C_0, C_1, C_2, \Delta t$	coefficients of the polynomial fit of the bistatic range history of a channel
$\Delta C_{0,i}(x)$	topographic residual of channel i: C_0 of a scatterer at height $h(x)$ minus C_0 of the reference point
X	ground distance between parcels folded onto the same Doppler bin, $X \approx r_0 \lambda \mathrm{PRF}_{\mathrm{op}}/(2v)$
δx	resolution of one SATA sub-aperture, $\sqrt{\lambda r_0/2}$
$H_i(f)$	transfer function of channel i relative to the monostatic reference
$g(x)$	focused full-rate azimuth line (unknown of the joint reconstruction)
μ	Tikhonov regularisation parameter

Acronym	
CGLS	conjugate gradients for least squares
DEM	digital elevation model
DPCA	displaced phase centre antenna (uniform effective sampling)
HRWS	high-resolution wide-swath
IRF	impulse response function
SATA	sub-aperture topography- and aperture-dependent algorithm

1 Introduction

1.1 Context

A conventional SAR cannot improve its azimuth resolution and its swath width at the same time: a fine azimuth resolution needs a short antenna and therefore a wide Doppler bandwidth, which must be sampled with a high PRF, while a wide swath needs a long interval between pulses. Multichannel systems break this trade-off by receiving each pulse with several apertures displaced along track [1, 2]. Each channel is sampled at a PRF below the Doppler bandwidth and is therefore aliased, but the set of channels samples the azimuth signal at the full rate, and the unambiguous spectrum is recovered by an *azimuth reconstruction*: a bank of filters that combines the channels so that the aliased parts cancel.

Multistatic formations distribute the receivers over several platforms. They lift the limit set by the length of a single antenna and offer flexibility, but the baselines between platforms are no longer small [3, 4]. In particular, a *cross-track* baseline makes the bistatic range history of a scatterer depend on its height. The reconstruction filters are derived for a reference geometry (here: the reference point of the scene at height $h_0 = 0$); a scatterer at another height carries a residual phase that is different in each channel. The filters then no longer match the data, the aliased parts no longer cancel and azimuth ambiguities appear in the image.

The same kind of height-dependent mismatch is known from airborne motion compensation, where the compensation is computed for a reference height and leaves residual errors for the rest of the topography. Prats et al. proposed the sub-aperture topography- and aperture-dependent (SATA) algorithm to correct it with an external DEM [5, 6]: the azimuth signal is split into short sub-apertures, each Doppler frequency of a sub-aperture is associated with a ground position, and the phase residual of that position, read from the DEM, is removed.

1.2 Objective

The fellowship topic is to investigate whether SATA can correct the azimuth ambiguities caused by topography and cross-track baselines in multistatic SAR systems. During the work the question was made precise as:

1. How large are the ambiguities caused by unaccounted topography, as a function of the terrain, the baselines and the number of channels?
2. How much of them can be removed using *only the system parameters and a DEM*, with no knowledge of where the scatterers are?

The second condition is strict on purpose. Several intermediate methods tested during the project reached the reference level only because their correction maps were built from the positions of the simulated targets; such methods are not applicable to real data and are not counted as solutions here (Section 4.2.3).

1.3 Contributions

- A numerical simulator of the multistatic acquisition and of the multichannel reconstruction in one dimension (azimuth) and in two dimensions (azimuth and range, 3-D scenes), with along- and cross-track baselines and arbitrary terrain (Chapters 2 and 3).
- An analysis of SATA applied to each aliased channel before the reconstruction, showing when it is exact and predicting the residual it leaves on general terrain (Chapter 4).
- A joint reconstruction that places the DEM phase inside the inversion, first per sub-aperture and finally over the whole line, solved by regularised least squares with conjugate gradients (Chapter 5), and its 2-D extension (Chapter 6).
- Systematic sweeps over terrains, numbers of channels and baselines, and an audit of every method with a DEM only (Chapter 7).

1.4 Outline

Chapter 2 introduces the acquisition geometry and the topographic residual ΔC_0. Chapter 3 reviews the multichannel reconstruction and explains how topography produces ambiguities. Chapter 4 describes SATA and its limit for aliased channels. Chapter 5 presents the joint reconstruction and Chapter 6 its 2-D version. Chapter 7 collects the results and Chapter 8 concludes. Appendix A maps the methods to the code in the repository.

2 Acquisition geometry and signal model

2.1 Formation and parameters

One platform transmits and receives (channel 0); the other $N_{\mathrm{rx}} - 1$ receivers fly behind it. Receiver i is displaced by $b_{\mathrm{at},i}$ along track and by $b_{\mathrm{xt},i}$ across track (horizontally, towards the scene) relative to the transmitter (Figure 2.1). The along-track baselines satisfy the displaced phase centre (DPCA) condition (Figure 2.2), so that the effective phase centres of the N_{rx} channels are uniformly spaced and the multichannel sampling is uniform at the full PRF; the study is about the cross-track baselines, which are drawn at random in $[0, {b_{\mathrm{xt}}}^{\max}]$ (with the largest equal to ${b_{\mathrm{xt}}}^{\max}$) or taken from fixed sets. Channel 0 is always the transmitter ($b_{\mathrm{at}} = b_{\mathrm{xt}} = 0$). The parameters are listed in Table 2.1.

Table 2.1: Parameters of the simulated system (`sata1d/config.py`).

Wavelength λ	0.25 m
Platform altitude H	720 km
Slant range of the reference point r_0	766.2 km
Look angle θ	20°
Platform velocity / effective velocity	7 688.5 m/s / 7 408.5 m/s
Antenna length	12 m
Processed Doppler bandwidth B_{a}	1 235 Hz
Full PRF	2 000 Hz
Channel PRF $\mathrm{PRF}_{\mathrm{op}} = \mathrm{PRF}/N_{\mathrm{rx}}$	1 000 Hz to 250 Hz ($N_{\mathrm{rx}} = 2$ to 8)
DPCA along-track spacing $2v/\mathrm{PRF}$	7.69 m
Cross-track baselines ${b_{\mathrm{xt}}}^{\max}$	100 m, 300 m and 500 m, and fixed sets

2.2 Signal model

The simulator works with point targets. In the 1-D study the targets lie on a terrain profile $h(x)$ along azimuth; each target is placed on the iso-range surface of the reference point (same monostatic slant range r_0), so that only its height changes the bistatic geometry. The reference point is at $x = 0$, $h = 0$. In the 2-D study (Chapter 6) the terrain is a surface $h(x, y)$ and the data are range compressed. For channel i the two-way path of a scatterer, $R_{\mathrm{T}}(t) + R_{\mathrm{R},i}(t)$, is fitted by a polynomial around the time of closest approach,

$$R_i(t) \approx C_{0,i} + C_{1,i}\,(t - \Delta t_i) + C_{2,i}\,(t - \Delta t_i)^2, \tag{2.1}$$

and the azimuth signal of the channel is

$$s_i(t) = \sum_p a_p\, w(t - t_p)\, \exp\!\left(-\mathrm{j}\frac{2\pi}{\lambda} R_{i,p}(t)\right), \tag{2.2}$$

with a_p the complex amplitude of scatterer p and w the antenna pattern. The reconstruction filters (Chapter 3) are built from the coefficients $C_{0,i}, C_{1,i}, C_{2,i}, \Delta t_i$ of the *reference point*.

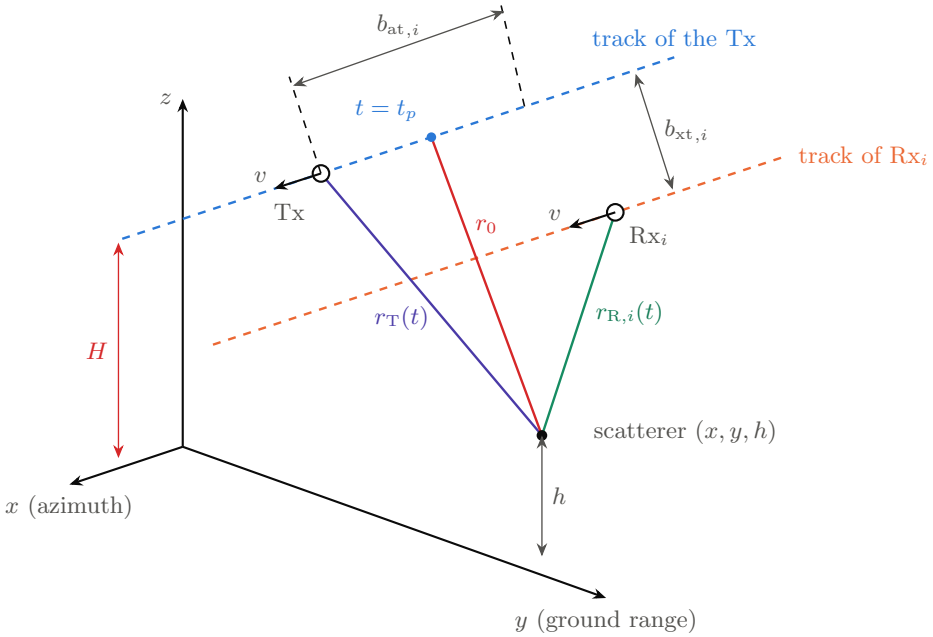


Figure 2.1: Geometry of one channel. The transmitter (Tx) and receiver i (Rx$_i$) fly on parallel tracks at the same altitude H, separated by $b_{\mathrm{at},i}$ along track and $b_{\mathrm{xt},i}$ across track. The echo of channel i travels $r_{\mathrm{T}}(t) + r_{\mathrm{R},i}(t)$. The time of closest approach t_p and the range r_0 fix the position of the scatterer on the line; its height h is what the topography adds. Not to scale.

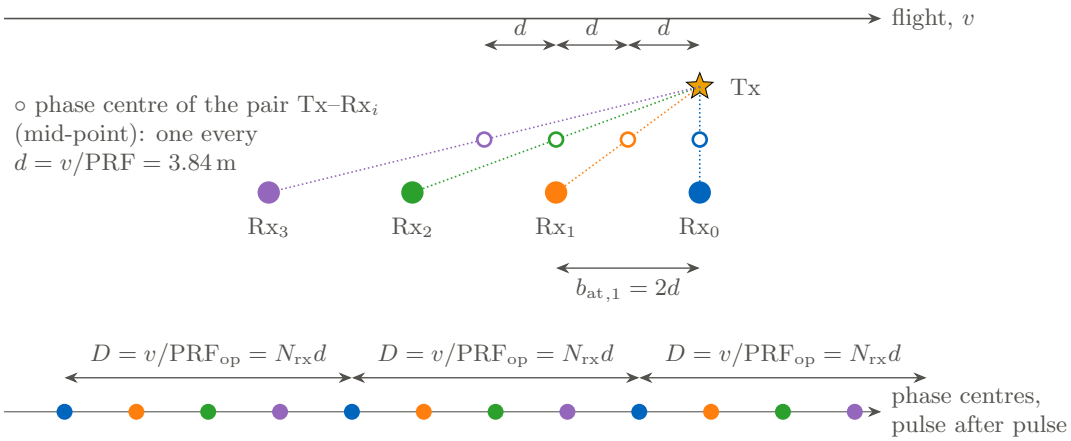


Figure 2.2: DPCA arrangement seen from above ($N_{\mathrm{rx}} = 4$). Receiver i trails the transmitter by $b_{\mathrm{at},i} = 2id$, so the phase centre of the pair (the mid-point) trails it by id. At every pulse the four phase centres occupy four equally spaced positions; between pulses the platform moves $D = N_{\mathrm{rx}}d$ and the pattern repeats. Together the channels sample the ground every d, like a single channel at the full PRF.

2.3 The topographic residual ΔC_0

For a scatterer at height h the coefficient C_0 of channel i differs from the one of the reference point. The difference is the *topographic residual*

$$\Delta C_{0,i}(h) = C_{0,i}(\text{scatterer}) - C_{0,i}(\text{reference point}) \approx -\frac{b_{\text{xt},i}\, h}{r_0 \tan\theta}, \qquad \varphi_i(h) = \frac{2\pi}{\lambda}\Delta C_{0,i}(h), \quad (2.3)$$

i.e. the interferometric phase of the cross-track baseline; it vanishes for the transmitter channel ($b_{\text{xt}} = 0$), since the targets are on the iso-range surface. The simulator computes $\Delta C_{0,i}$ exactly from the range histories (`residual_C0`); the approximation (2.3) agrees within a fraction of a millimetre for the cases of this report. For a scatterer at 500 m and $b_{\text{xt}} = [63.7, 27.0, 4.1, 1.7]$ m the residuals are $[-113.9, -48.2, -7.3, -3.0]$ mm, i.e. up to half a wavelength. The other coefficients change much less with height: the topography acts as a channel-dependent *constant* phase over the aperture of each scatterer, a fact used by the joint reconstruction (Chapter 5).

Figure 2.3 shows where (2.3) comes from. Moving from the reference point to a scatterer at height h on the same range arc of the transmitter is a displacement $s = h/\sin\theta$ perpendicular to the line of sight of the transmitter, which does not change its path. The line of sight of receiver i is rotated by $\Delta\theta = b_{\text{xt},i}\cos\theta/r_0$, so its path changes by $s\,\Delta\theta = b_{\text{xt},i}h/(r_0\tan\theta)$.

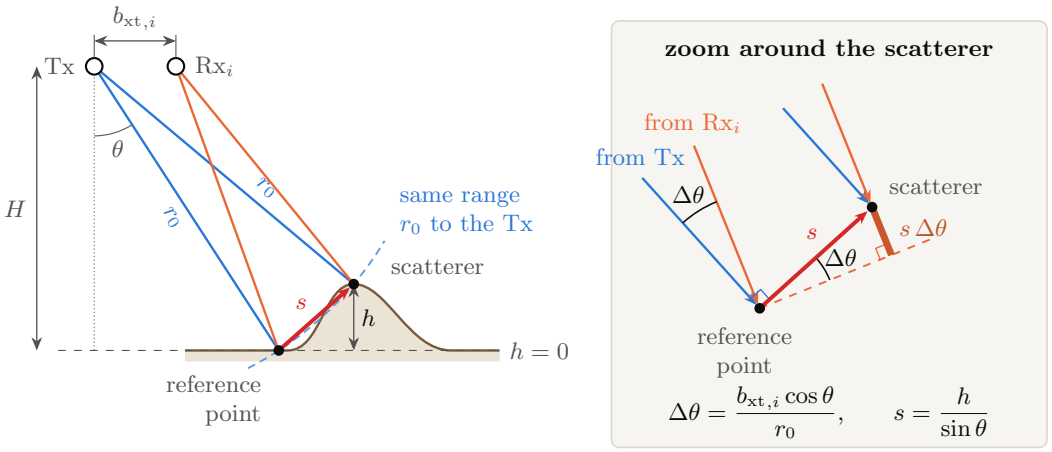


Figure 2.3: Left: plane perpendicular to the track (not to scale). The reference point and the scatterer are at the same range r_0 from the transmitter, at different heights: climbing h along the arc is a displacement $s = h/\sin\theta$. Right: zoom around the scatterer. Since $s \ll r_0$ the lines of sight of each platform to the two points are parallel; s is perpendicular to the line of sight of the transmitter, but the one of the receiver is rotated by $\Delta\theta$, and its path becomes $s\,\Delta\theta$ shorter (thick orange segment).

The sensitivity is $\partial\varphi_i/\partial h = 2\pi b_{\text{xt},i}/(\lambda r_0 \tan\theta) \approx 9.0 \times 10^{-5}\, b_{\text{xt},i}$ rad/m, which also sets the tolerance to DEM errors (Section 7.4).

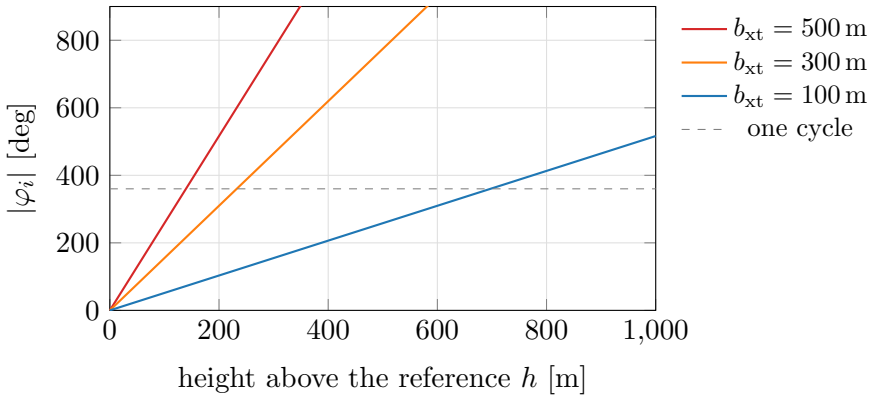


Figure 2.4: Phase of the topographic residual (2.3) against height. With $b_{\mathrm{xt}} = 300\,\mathrm{m}$ the sensitivity is 1.55° per metre of height: a 230 m hill already produces a full cycle.

3 Multichannel azimuth reconstruction and topography

3.1 Aliased channels and folded parcels

Each channel is sampled at $\mathrm{PRF}_{\mathrm{op}} = \mathrm{PRF}/N_{\mathrm{rx}}$, below the Doppler bandwidth $B_{\mathrm{a}} = 1\,235\,\mathrm{Hz}$. In the Doppler domain of the channel, every frequency $\phi \in [-\mathrm{PRF}_{\mathrm{op}}/2, \mathrm{PRF}_{\mathrm{op}}/2)$ is the sum of N_{rx} *parcels* of the full-rate spectrum U, taken at

$$f_q = \phi + q\,\mathrm{PRF}_{\mathrm{op}} - \tfrac{N_{\mathrm{rx}}-1}{2}\mathrm{PRF}_{\mathrm{op}}, \qquad q = 0, \ldots, N_{\mathrm{rx}} - 1, \tag{3.1}$$

each weighted by the transfer function H_i of the channel relative to the monostatic reference:

$$S_i(\phi) = \sum_q H_i(f_q)\, U(f_q) \quad \Longleftrightarrow \quad \boldsymbol{s}(\phi) = \mathbf{H}(\phi)\, \boldsymbol{u}(\phi), \tag{3.2}$$

with $\boldsymbol{s}$ the N_{rx} channel spectra and $\boldsymbol{u}$ the N_{rx} parcels. The standard reconstruction [1, 2] inverts this square system at every ϕ, $\hat{\boldsymbol{u}} = \mathbf{P}(\phi)\boldsymbol{s}(\phi)$ with $\mathbf{P} = \mathbf{H}^{-1}$, and places the parcels at their true Doppler frequency to obtain the full-rate signal. $\mathbf{H}$ is built from the range-history coefficients of the reference point (`GetCoeffNu`, `GetInversionFilters`).

Through the Doppler–angle relation, a Doppler frequency f seen from the platform at azimuth position x_c corresponds to the ground position

$$x(f) = x_c + r_0 \tan\!\big(\arcsin(\lambda f/2v)\big) \approx x_c + \frac{\lambda r_0}{2v}\, f, \tag{3.3}$$

which we call the *ruler*. Parcels one $\mathrm{PRF}_{\mathrm{op}}$ apart therefore come from ground positions

$$X \approx \frac{\lambda r_0\, \mathrm{PRF}_{\mathrm{op}}}{2v} \qquad (6.23\,\mathrm{km} \text{ for } \mathrm{PRF}_{\mathrm{op}} = 500\,\mathrm{Hz}), \tag{3.4}$$

apart, i.e. 0.81 s of azimuth time. This distance is the key length of the problem: one bin of an aliased channel mixes the echoes of the terrain at x, $x \pm X$, $x \pm 2X, \ldots$ (Figure 3.1).

3.2 How topography creates ambiguities

For a scatterer at height h, channel i carries the extra constant phase $\mathrm{e}^{-\mathrm{j}\varphi_i}$ of (2.3), so the data are $\boldsymbol{s} = \mathbf{D}\mathbf{H}\boldsymbol{u}$ with $\mathbf{D} = \mathrm{diag}(\mathrm{e}^{-\mathrm{j}\varphi_i})$, and the standard reconstruction returns

$$\hat{\boldsymbol{u}} = \mathbf{H}^{-1}\mathbf{D}\,\mathbf{H}\,\boldsymbol{u} = \mathbf{L}\,\boldsymbol{u}. \tag{3.5}$$

In the DPCA configuration $\mathbf{H}$ is, up to diagonal factors, a discrete Fourier matrix over the channels, and the leakage matrix becomes Toeplitz,

$$L_{q'q} = c_{q'-q}, \qquad c_m = \frac{1}{N_{\mathrm{rx}}} \sum_{i=0}^{N_{\mathrm{rx}}-1} \exp\!\Big[\mathrm{j}\Big(\frac{2\pi i m}{N_{\mathrm{rx}}} - \varphi_i\Big)\Big]. \tag{3.6}$$

c_0 is the gain of the true parcel and c_m, $m \neq 0$, the amplitude that parcel q leaks into parcel $q + m$, which appears in the image as an ambiguity displaced by mX ($0.81\,\mathrm{s}\cdot m$). If all channels

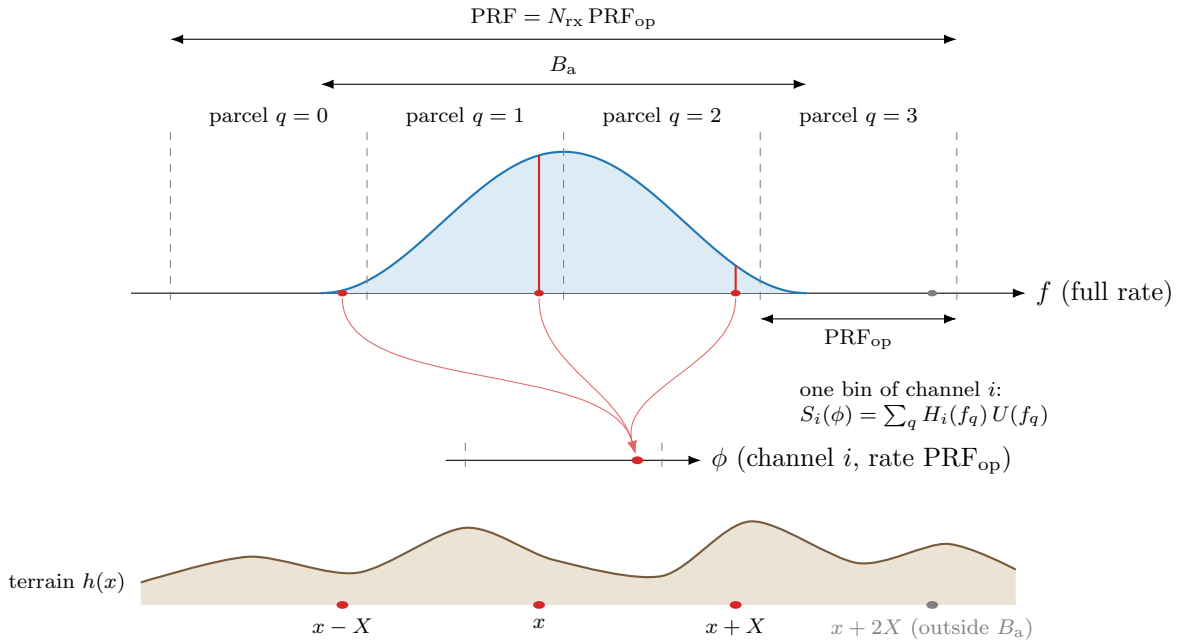


Figure 3.1: Folding of the azimuth spectrum in an aliased channel ($N_{\mathrm{rx}} = 4$). The four parcels of one channel bin come, through the ruler (3.3), from ground positions X apart, which in general have different heights.

have the same phase, $c_m = 0$ for $m \neq 0$ and the reconstruction is exact; with cross-track baselines the phases differ and the ambiguity level grows with the spread of $b_{\mathrm{xt},i}h$. Weighted by the fraction of the target bandwidth that falls in the leaking parcels, $|c_m|$ predicts the ambiguity of the uncorrected reconstruction within 0.15 dB for heights above 50 m.

Two consequences recur in the rest of the report:

- The amplitudes c_m are the discrete Fourier transform of the channel phases. Baselines for which several φ_i coincide modulo 2π, or that are equally spaced, make the system badly conditioned at certain heights: the channels no longer separate the parcels.
- With two channels there is no redundancy at all: two parcels, two equations.

3.3 Ambiguity metric

All results are reported as the *highest ambiguity*: the largest value of the azimuth impulse response, in dB relative to the peak of the reference, inside ±20 ms of the ambiguity positions of every target (±1 and ±2 times the ambiguity distance of the channel PRF). The reference is the monostatic full-rate signal of the same scene, processed with the same band limitation; its own value (sidelobes of neighbouring targets, band edge) is the floor that any reconstruction can reach. Repeating the measurement with orders up to ±5 for six and eight channels did not change the conclusions.

4 SATA before the reconstruction

4.1 The SATA algorithm

SATA was developed for airborne motion compensation [5, 6]. Its idea is that a short piece of the azimuth signal is a low-resolution image of the ground: inside a sub-aperture, each Doppler frequency belongs to one ground position through the ruler (3.3). The topography-dependent phase can therefore be removed frequency by frequency, with the value of the DEM at the position that frequency points to. The steps, for one channel, are (Figure 4.1):

1. cut the azimuth line into sub-apertures with triangular windows and 50 % overlap;
2. Fourier transform each sub-aperture (with zero padding);
3. for every bin f, compute the ground position $x(f)$ with the ruler and multiply the bin by $\exp\big(+\mathrm{j}\,2\pi\Delta C_{0,i}(x(f))/\lambda\big)$, with $\Delta C_{0,i}$ read from the DEM;
4. inverse transform and overlap-add.

The length of the sub-aperture balances two errors: it must be short enough that the frequency-to-position mapping is sharp, and long enough that the frequency resolution resolves the ground. The optimum gives a ground resolution $\delta x = \sqrt{\lambda r_0/2} = 309.5\,\mathrm{m}$; at $\mathrm{PRF}_{\mathrm{op}} = 500\,\mathrm{Hz}$ this is a block of 20 samples (40 ms), a hop of 10 samples and a zero-padded FFT of 128 points.

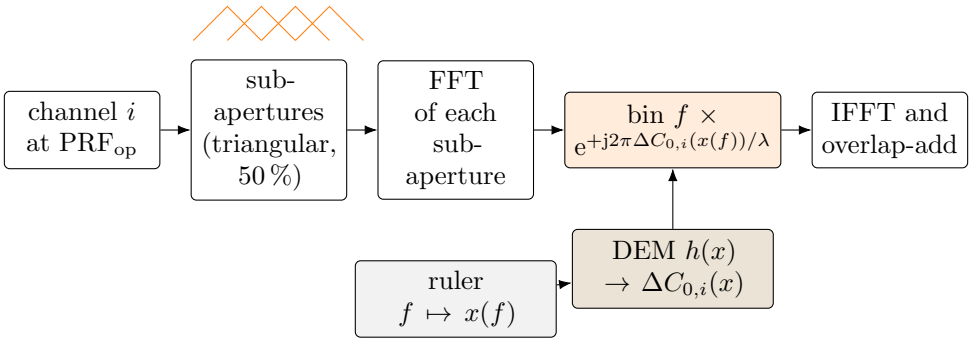


Figure 4.1: The SATA kernel applied to one channel. The only external information is the DEM; the ruler associates each sub-aperture bin with a ground position.

4.2 SATA on each channel, then the standard reconstruction

The natural way to use SATA in a multichannel system is to correct every channel with its own $\Delta C_{0,i}$ and then run the standard reconstruction. Two variants were implemented:

Whole band each bin of the channel is labelled with the position of its base-band parcel; the corrected channels go through the standard reconstruction.

Per sub-band for every output sub-band k, each bin is labelled with the position of the parcel of sub-band k (absolute Doppler), and sub-band k of the output is reconstructed only from the channels corrected for k.

Both variants are the identity with a zero map ($< 10^{-15}$), rotate the line by the right phase with a constant map, and the kernel agrees with the reference routine `sata_for_topography` within 0.44 % rms.

4.2.1 Why it is exact only on special terrain

The channel is aliased: one bin holds the parcels q of the positions $x_q = x_b + \rho f_q$, X apart (Figure 3.1). SATA multiplies the whole bin by a *single* phase, the one of the labelled position x_r. After the correction, parcel q keeps the error

$$\varepsilon_{iq} = \varphi_i(x_q) - \varphi_i(x_r) = -\frac{2\pi\, b_{\mathrm{xt},i}\,\big[h(x_q) - h(x_r)\big]}{\lambda\, r_0 \tan\theta}, \tag{4.1}$$

and the standard reconstruction returns $\hat{\boldsymbol{u}} = \mathbf{L}\boldsymbol{u}$ with $\mathbf{L} = \mathbf{H}^{-1}\mathrm{diag}(\mathrm{e}^{-\mathrm{j}\varepsilon})\mathbf{H}$: the same leakage structure as (3.6), now driven by the *height differences between folded positions*. Hence:

- $\mathbf{L} = \mathbf{I}$, whatever the baselines, only if $h(x) = h(x \pm X) = h(x \pm 2X) = \ldots$: flat terrain, or a terrain periodic with period X;
- for two channels, a -30 dB ambiguity tolerates only about 7 m of height difference between folded positions with $b_{\mathrm{xt}} = 100$ m, and 1.4 m with 500 m;
- in the whole-band variant only the fraction of the target spectrum that falls in the labelled parcel receives the phase of its own position (about 78 %, 52 %, 39 %, 26 % and 20 % for $N_{\mathrm{rx}} = 2, 3, 4, 6, 8$); in the per-sub-band variant the output parcel is right but all the others leak into it.

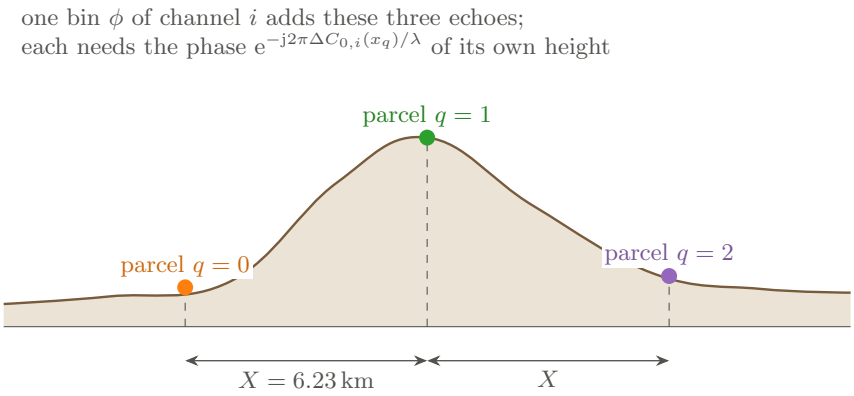


Figure 4.2: The conflict of one bin. Its parcels come from places X apart, with different heights. SATA on the channel applies one phase to the whole bin; the joint reconstruction applies the phase of each place to its own parcel, inside the system of equations.

Correcting each parcel needs the parcels separated, and separating them needs the phases corrected: the two steps cannot be done one after the other. This is what the joint reconstruction of Chapter 5 does in a single system.

4.2.2 Verification

The analysis was checked in three ways (83 automatic checks in `joint_cases/audit_checks.py`, all passing):

- **Positive control.** On the terrain $h = 100 - 100\cos(2\pi x/X)$, whose period is exactly X, whole band, per sub-band, joint and oracle coincide (-62.7 dB, -44.2 dB and -34.0 dB for 1, 5 and 25 targets).
- **Negative control.** The same terrain with period $1.5X$ or $2X$: whole band -32 dB to -27 dB, per sub-band -26 dB to -23 dB, while the joint reconstruction stays at the oracle.
- **Prediction.** The level left by SATA, computed only from the fold addresses and the DEM, matches the simulation within 1.3 dB rms (whole band) and 0.7 dB rms (per sub-band) over 680 runs. The residual is structural, not an implementation error.

SATA can even make things worse: a single target at $h = 0$ on a terrain of period 50 km ($b_{\mathrm{xt}} \leq 100$ m) reaches -63.7 dB without correction, but -36.3 dB after SATA whole band and -27.8 dB per sub-band, because the neighbouring terrain folded onto the same bins is not at $h = 0$.

A complementary run removes the aliasing (each channel at $\mathrm{PRF}_{\mathrm{op}} = 2\,000$ Hz $> B_{\mathrm{a}}$): there SATA alone brings smooth terrains close to the reference (peak of the worst target within 0.3 dB to 1.4 dB) and is limited only on steep terrain by its 310 m cell. The limit discussed above is therefore a consequence of the aliasing, not of SATA itself.

4.2.3 Correction maps that use the target positions

During the project, maps were also built from the simulated targets: the ΔC_0 of each target written at its cell and copied at the folded positions $\pm X$ (*block map with copies*), or the value of the nearest target. With such maps SATA per channel reaches the oracle for isolated targets, because the map then encodes exactly which parcel is occupied. A real scene provides no such information. These maps are kept in the code for reference but are *not* part of any method evaluated in Chapter 7; every method there receives only the system parameters and the DEM, and removing the target list from the configuration given to the methods does not change a single bit of their output.

5 Joint topography correction and reconstruction

The analysis of Section 4.2.1 shows that the correct phase (the ΔC_0 read from the DEM) is already known; what fails is applying it per channel before the parcels are separated. The joint reconstruction moves the DEM phase inside the inversion.

5.1 Joint reconstruction in the sub-apertures

With the same sub-apertures as SATA, for every sub-aperture and every bin ϕ of the N_{rx} channel spectra the model (3.2) is written with the phase of each parcel at its own ground position:

$$S_i(\phi) = \sum_q H_i(f_q)\, \exp\!\Big(-\mathrm{j}\frac{2\pi}{\lambda}\Delta C_{0,i}(x_q)\Big)\, U_q, \qquad x_q = x_c + \frac{\lambda r_0}{2v} f_q, \tag{5.1}$$

and solved for the parcels U_q, which are then placed at their true Doppler, transformed back and overlap-added at the full PRF (`reconstruct_joint`). The steps are summarised in Figure 5.1. With $\Delta C_0 = 0$ this is the standard reconstruction ($\sim 10^{-7}$); with one target it equals the oracle. Its limit is the resolution of the sub-aperture: the DEM is read once per bin of a 40 ms block, i.e. the terrain phase is taken as constant over about ± 600 m of ground. A slope s with a cross-track baseline is equivalent to shifting the phase centre of the channel by $b_{\mathrm{xt}} s/(2\tan\theta)$; when that shift is not small compared with the block, the per-bin model breaks. With $\Delta\varphi = \frac{2\pi}{\lambda}\frac{b_{\mathrm{xt}}{}^{\max}\, s\, \delta x}{r_0 \tan\theta}$ the phase change inside one SATA cell, the method stays within about 10 dB of the reference for $\Delta\varphi \lesssim 0.5$ rad and fails by 10 to 35 dB above 1 rad.

5.2 Joint reconstruction over the whole line

The final version removes the sub-apertures. The topographic residual of a scatterer is a constant phase over its whole aperture (Chapter 2); in the *focused* full-rate line $g(x)$ it is therefore a pointwise multiplication. The channels are modelled as

$$\boldsymbol{s}_i = \mathcal{D}\Big\{k_i * \big[\, g(x)\, \mathrm{e}^{-\mathrm{j}\varphi_i(x)}\big]\Big\}, \qquad \varphi_i(x) = \frac{2\pi}{\lambda}\Delta C_{0,i}\big(h(x)\big), \tag{5.2}$$

where $\varphi_i(x)$ comes from the DEM only (`line_maps`), k_i is the response of channel i to the reference point (the azimuth chirp filtered by the reference-point filter H_i of the standard reconstruction) and $\mathcal{D}$ is the decimation to $\mathrm{PRF}_{\mathrm{op}}$ (Figure 5.2). Writing (5.2) as $\boldsymbol{s} = \mathbf{A}\boldsymbol{g}$, the line is estimated by Tikhonov-regularised least squares,

$$\hat{\boldsymbol{g}} = \arg\min_{\boldsymbol{g}} \sum_i \big\|\boldsymbol{s}_i - \mathbf{A}_i\boldsymbol{g}\big\|^2 + \mu^2\|\boldsymbol{g}\|^2, \tag{5.3}$$

solved with conjugate gradients for least squares (CGLS) [7, 8]. Every iteration costs one application of $\mathbf{A}$ and of its adjoint $\mathbf{A}^{\mathsf{H}}$ (two FFTs per channel each). The regularisation is $\mu = 0.03\,\sigma_{\max}$, with $\sigma_{\max}$ estimated by a few power iterations, and the problem is solved in two passes (iterated Tikhonov: the second pass solves for what the first one left), which turns the filter factor $\sigma^2/(\sigma^2+\mu^2)$ into $1-(1-\sigma^2/(\sigma^2+\mu^2))^2$. Each pass runs up to 60 iterations. The output is re-chirped with the reference response, $U = k * \hat{g}$, so that it has the same form as the standard reconstruction. The complete algorithm is in Figure 5.3.

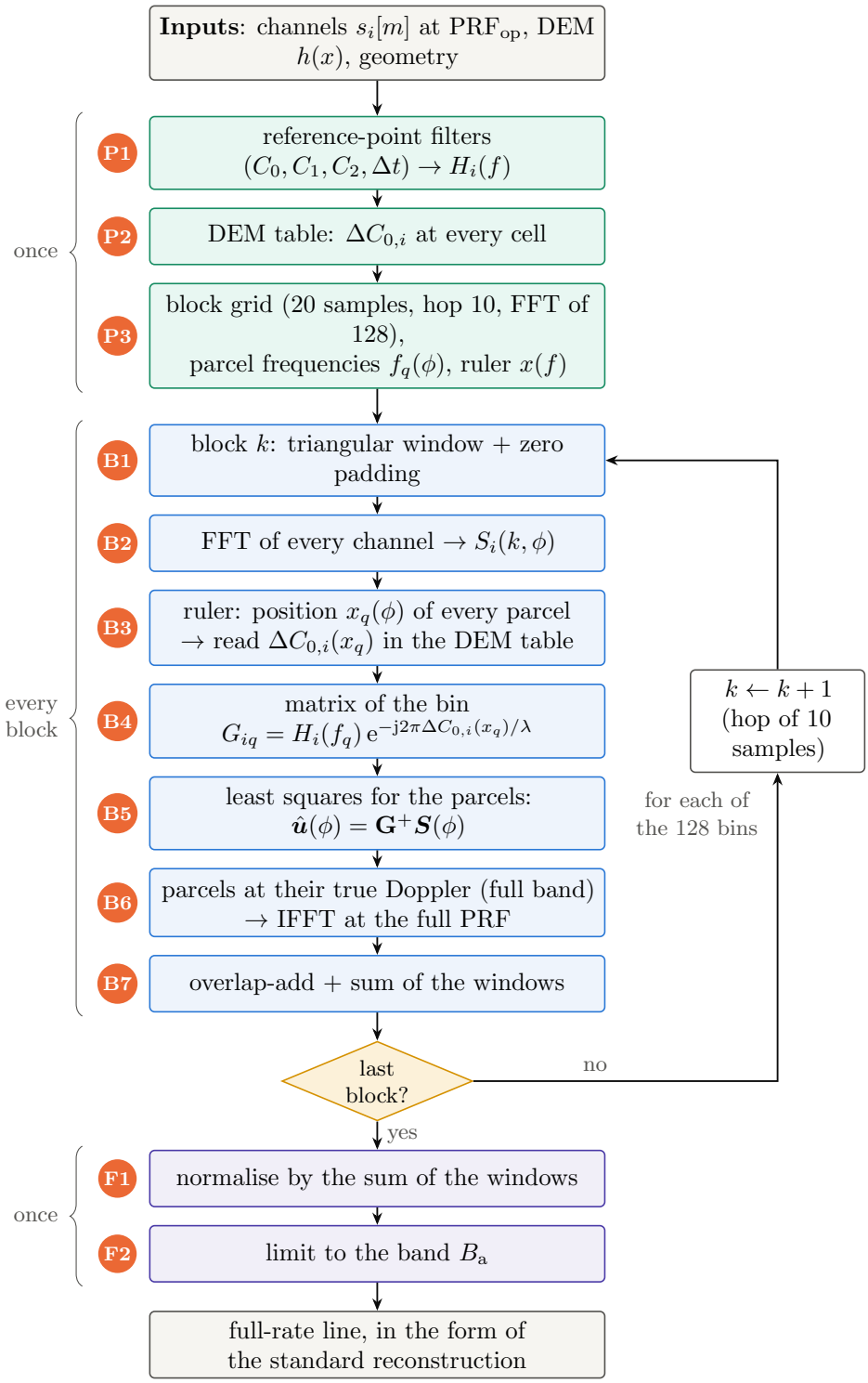


Figure 5.1: The joint reconstruction in the sub-apertures, as implemented in `reconstruct_joint`. Green: preparation, done once. Blue: the loop over the blocks of 20 samples; inside each block a small least-squares system is solved for every bin. Violet: finishing. The DEM is read once per bin, at the position the ruler gives to each parcel.

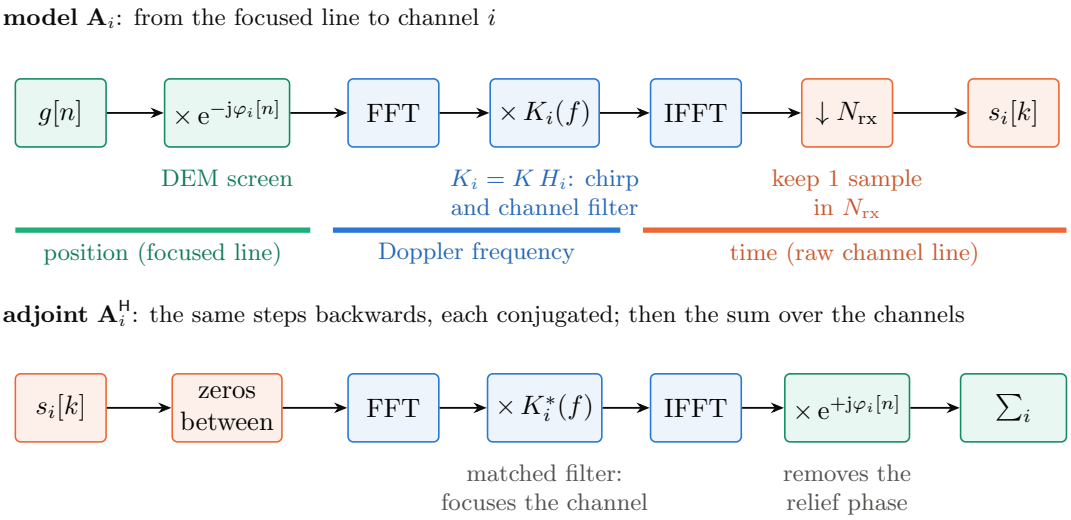


Figure 5.2: The model $\mathbf{A}$ and its adjoint. Each operation is done in the domain where it is simple: the relief phase is a multiplication in *position*, the chirp and the channel filter a multiplication in *Doppler frequency*, the sampling a selection of samples in *time*; the FFTs only move between the domains. The adjoint is the usual SAR processor: it focuses each channel, removes its relief phase and sums the channels.

5.3 Properties

- **Consistency.** With a zero DEM the method reproduces the standard reconstruction within 0.1 % rms.
- **Information used.** Only the channels, the system geometry and the DEM. The positions of the targets are never used (Section 4.2.3).
- **Self-test of the DEM.** The part of the channels that the model does not explain is −41 dB to −39 dB with the right DEM and −20 dB to −7 dB with a wrong one (mirrored or shifted by 1 km), which makes it a natural alarm for DEM errors.
- **Cost.** About 3.3 s per line with four channels in the 1-D simulator (median over 680 runs), against 0.15 s for SATA whole band.
- **Ill-conditioned configurations.** Where the baselines do not separate the parcels (two channels, equally spaced baselines, certain heights with long baselines) the information is not in the data; the regularisation then leaves the ambiguity instead of amplifying noise.

Figure 5.5 shows the example used throughout the development: five targets on a 1 000 m Gaussian mountain with $b_{xt} = [0, 134.9, 20.5, 8.3]$ m. The highest ambiguity goes from −8.7 dB without correction to −43.5 dB with the joint reconstruction; the reference is −48.7 dB, and −43.9 dB once it is limited to the processed band B_a, as every reconstruction is.

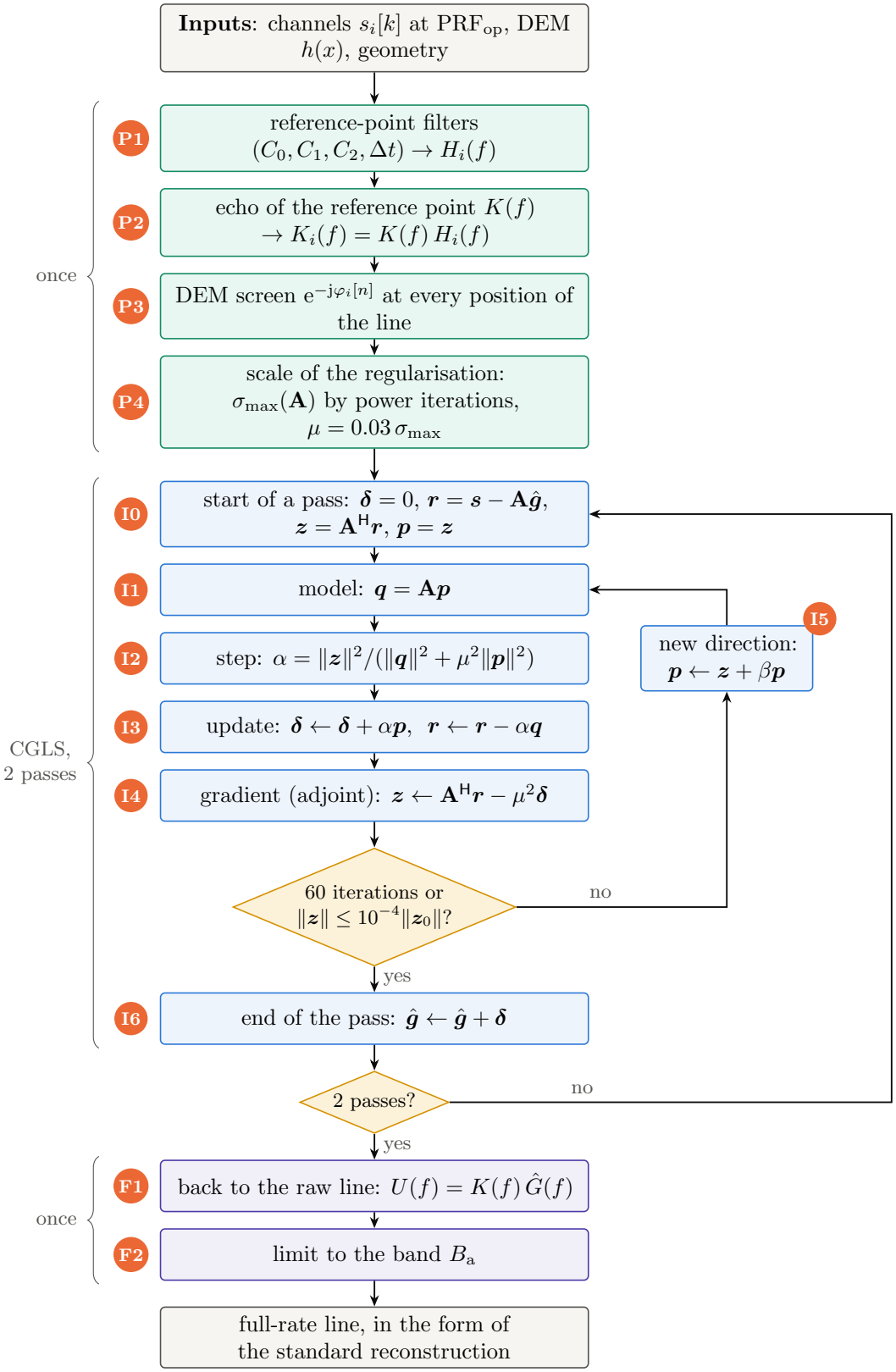
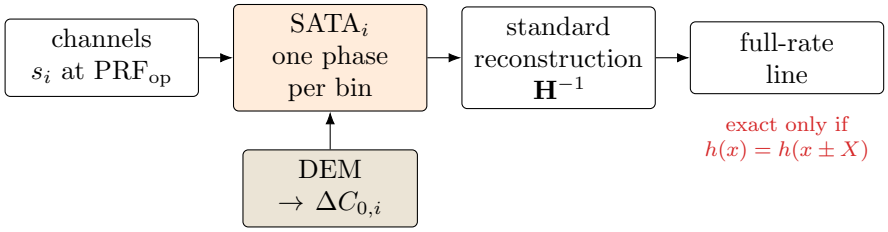


Figure 5.3: The joint reconstruction over the whole line, as implemented in `reconstruct_joint_line`. Green: preparation, done once. Blue: the CGLS iterations, in two passes (iterated Tikhonov); each iteration applies $\mathbf{A}$ and $\mathbf{A}^{\mathsf{H}}$ once (two FFTs per channel each). Violet: back to the raw full-rate line, in the same form as the standard reconstruction.

(a) SATA on each channel, then the standard reconstruction



(b) Joint reconstruction over the whole line (final version)

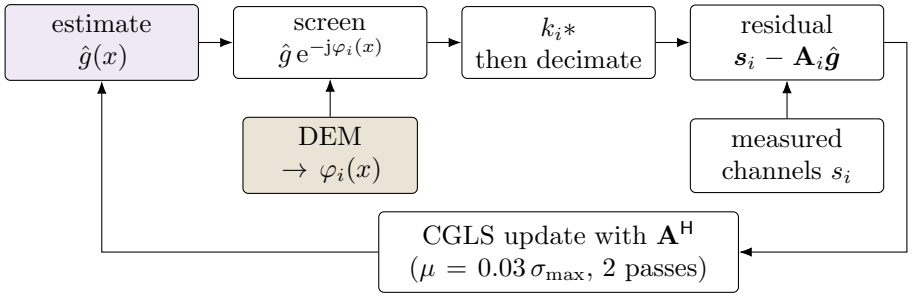


Figure 5.4: Where the DEM phase enters. (a) SATA gives each channel bin a single phase before the parcels are separated. (b) The joint reconstruction puts the phase of every ground position in the model of the channels and inverts model and topography together.

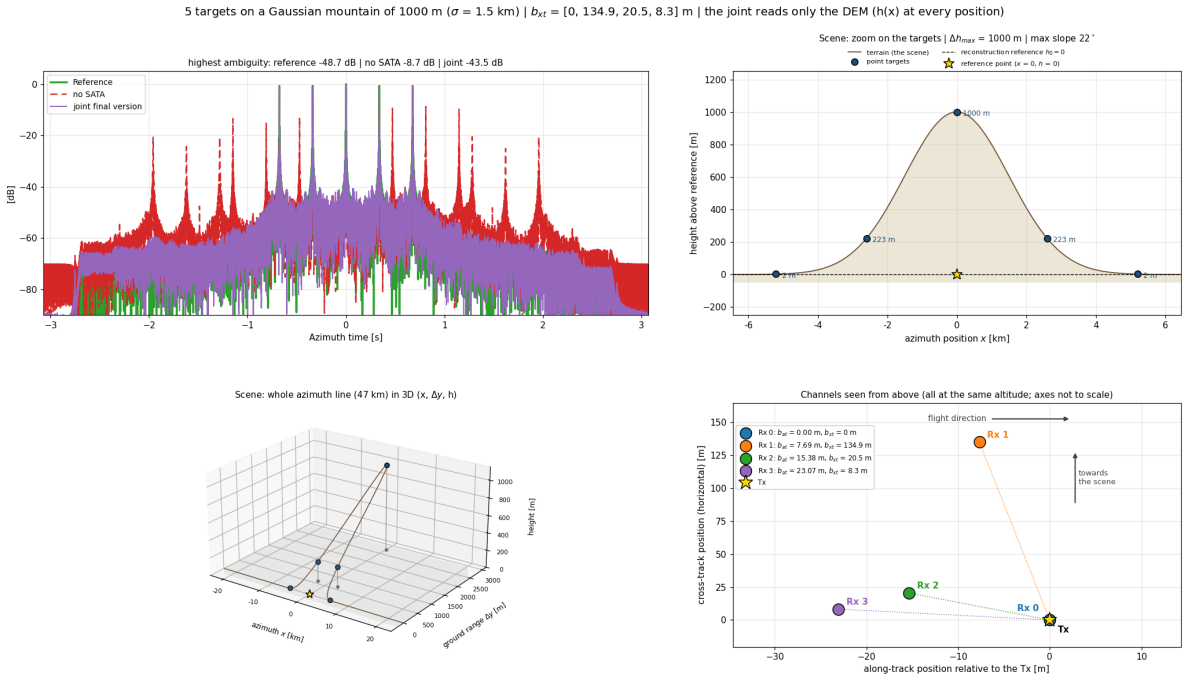


Figure 5.5: Five targets on a 1 000 m Gaussian mountain, four channels. Top left: azimuth impulse responses of the reference, of the standard reconstruction (no SATA) and of the joint reconstruction over the whole line. Right: the scene and the formation. The joint reconstruction reads only the DEM.

6 Extension to two dimensions

The 1-D study keeps every target at the slant range of the reference point. In a real scene the terrain varies in range as well, the residual $\Delta C_{0,i}$ depends on both coordinates, the echoes migrate in range and the range envelope of each channel is shifted by half its path difference. The `sata2d` package simulates 3-D scenes with along- and cross-track baselines and range-compressed data, and `sata2d/joint2d.py` extends the joint reconstruction to it.

6.1 Model

The unknown is the focused image $g[n, m]$ (Figure 6.2) on azimuth positions x_n (full PRF) and slant ranges of closest approach r_m. The range-compressed data of channel i are modelled as

$$\boldsymbol{s}_i = \mathcal{D}\Big\{ K_i \cdot \mathcal{F}_r\big[\, Q_i \cdot \mathcal{F}_a(S_i\, g)\big]\Big\} \;+\; (\text{envelope term}), \tag{6.1}$$

with

- S_i, the terrain screen at every pixel, $\exp\big(-\mathrm{j}2\pi\Delta C_{0,i}(x, r)/\lambda\big)$, computed from the DEM in radar geometry (the ground range of each pixel is found by a fixed-point iteration, Figure 6.1, which converges for range slopes below $\tan\theta$, i.e. without layover);
- $Q_i(f; r)$, what changes with the range of closest approach in the range-Doppler domain: the azimuth chirp relative to the reference range and the differences of the coefficients $C_0, C_1, C_2, \Delta t$ of the channel;
- $K_i(f, f_r)$, the 2-D kernel: the 2-D FFT of the simulated echo of the reference point in channel i at the full PRF (chirp, range migration, window of the pair, co-registration by $C_{0,i}/2$);
- the envelope term, the first-order shift of the range envelope by $\frac{1}{2}[\Delta C_{0,i} + C_{0,i}(r) - C_{0,i}(r_{\mathrm{ref}})]$, valid while this shift is small compared with the range resolution.

The image is kept on a range grid twice as fine as the data, because a phase that varies with range shifts the range spectrum. The adjoint passes the adjoint test to 2.6×10^{-7}; the solver is the same CGLS as in 1-D ($\mu = 0.03\,\sigma_{\max}$, two passes of up to 60 iterations).

6.2 The common range band

A cross-track baseline shifts the range spectrum seen by the channel, the spectral shift known from interferometry:

$$\nu_i = \frac{c\, b_{\mathrm{xt},i} \cos\theta}{2\lambda\, r \tan(\theta - \alpha)}, \tag{6.2}$$

with α the range slope of the terrain; on flat terrain this is 0.202 MHz per 100 m of baseline. Range frequencies that only some channels observe are not determined by the data (fewer equations than folded parcels), and the joint reconstruction leaves the full ambiguity there. In a control case (one target, flat terrain, $b_{\mathrm{xt}} = [0, 300, 119.8, 100.5]$ m, exact model) the full band reaches -40.5 dB, whereas the band common to all four channels reaches -66.1 dB, equal to the reference in the same band. The output is therefore limited, pixel by pixel, to the range frequencies that

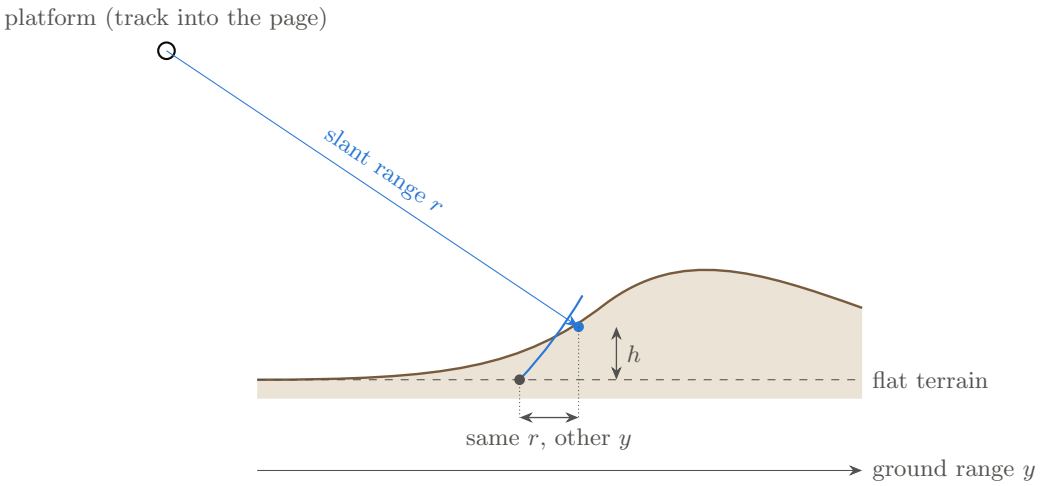
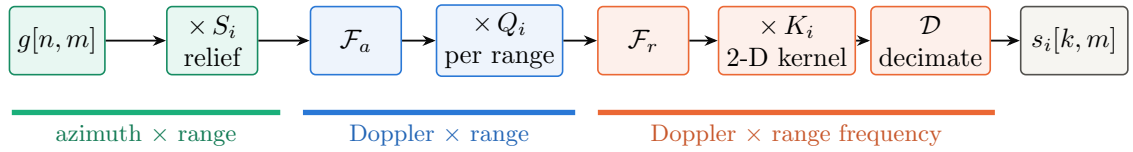


Figure 6.1: The DEM in radar geometry: a pixel at slant range r lies on the terrain at another ground range than on flat terrain. The ground range is found by the fixed-point iteration $y \leftarrow \sqrt{r^2 - [H - h(x,y)]^2}$, which contracts while the range slope stays below $\tan\theta$.

model: from the focused image to the range-compressed data of channel i



adjoint: repeat the spectrum, 2-D matched filter, remove the relief, sum the channels

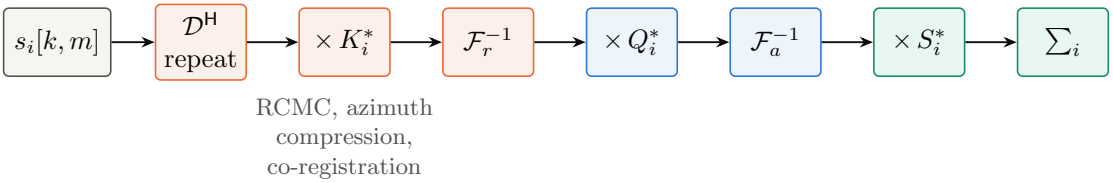


Figure 6.2: The model of the 2-D joint reconstruction and its adjoint (the envelope term is omitted). The relief acts per pixel, the range-dependent focusing per range line in the range-Doppler domain, and the 2-D kernel of the channel in the two-dimensional frequency domain.

every channel observes there, and the reference and the uncorrected reconstruction are compared in the same band. The price is range resolution, a fraction $b_{\mathrm{xt}}/{b_{\mathrm{xt}}}^{\mathrm{crit}}$ of the band; with the 5 MHz bandwidth of the sweeps the median pixel keeps 93 % of the band and the worst pixel 45 %. A wider system bandwidth (e.g. 50 MHz) would reduce this loss about tenfold.

Figure 6.3 shows a 3-D case: 49 targets on an isolated 1 000 m mountain with four channels and ${b_{\mathrm{xt}}}^{\mathrm{max}} = 300\,\mathrm{m}$.

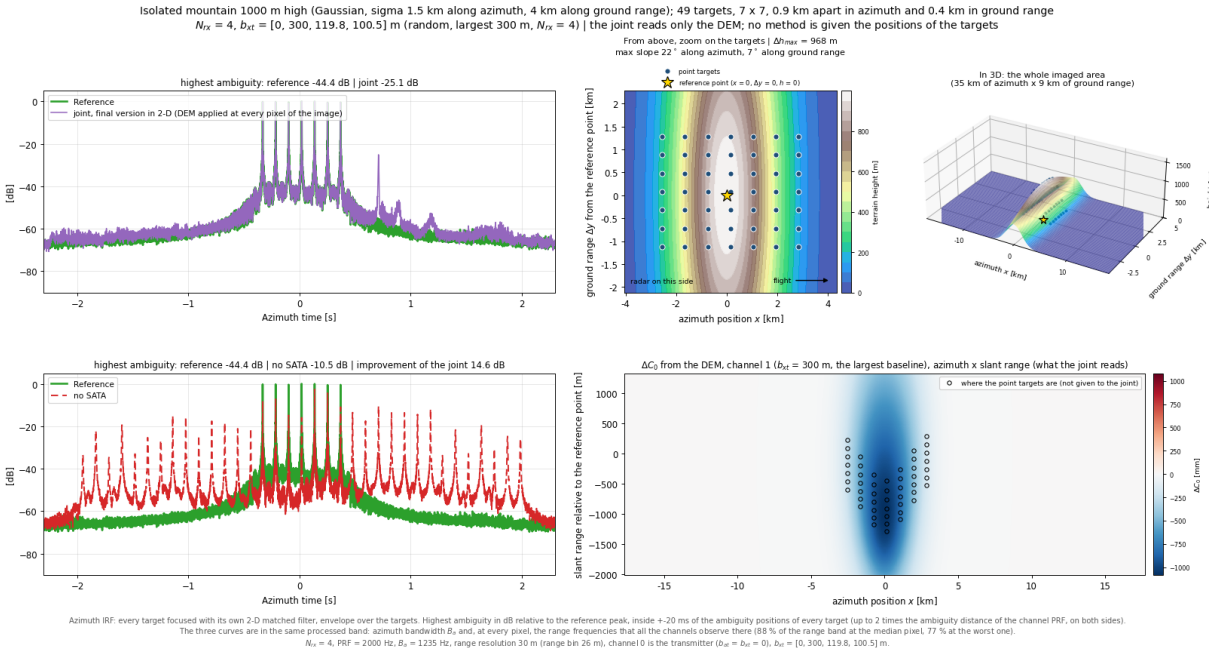


Figure 6.3: A page of the 2-D sweep: 49 targets on an isolated 1 000 m mountain (Gaussian, $\sigma = 1.5$ km in azimuth and 4 km in ground range), $N_{\mathrm{rx}} = 4$, $b_{\mathrm{xt}} = [0, 300, 119.8, 100.5]$ m. Top left: reference and joint reconstruction; bottom left: reference and no correction; right: the scene and the ΔC_0 of the channel with the largest baseline, which is all the joint reconstruction reads. This is one of the harder cases (the joint reconstruction leaves −25.1 dB, an improvement of 14.6 dB).

7 Results

All numbers in this chapter are the highest ambiguity of Section 3.3, in dB relative to the peak of the reference. Every method receives only the system parameters and the DEM. The scenes cover along-track ramps (3° to 20°), sines and cosines of several periods, and realistic terrains (hills, mountains, escarpments, plateaus, fractal alpine relief, glacial valleys, ridges, canyons, volcanoes), each with one, a few and many point targets (up to 49). The baselines are $N_{\mathrm{rx}} = 2, 3, 4, 6, 8$ with ${b_{\mathrm{xt}}}^{\max} = 100\,\mathrm{m}$, 300 m and 500 m (random, the largest exactly ${b_{\mathrm{xt}}}^{\max}$), plus five fixed four-channel sets: `u100` and `u500` (random), `s10` and `s50` (equally spaced, $[0, 10, 20, 30]$ and $[0, 50, 100, 150]$ m) and `pm50`. Channel 0 is always the transmitter.

7.1 All methods with a DEM only

Table 7.1 and Figure 7.1 compare the methods on 680 runs (34 scenes × 20 baseline sets; median reference −48.7 dB).

Table 7.1: All methods with the DEM only (680 runs, `joint_cases/audit_dem.py`).

Method	Median highest ambiguity [dB]	Runs within 10 dB of the reference [%]
No correction (standard reconstruction)	−8.7	3.5
SATA whole band, then reconstruction	−15.4	2.5
SATA per sub-band	−10.8	1.2
Joint reconstruction in the sub-apertures	−30.9	36.3
Joint reconstruction over the whole line	**−41.4**	**81.9**

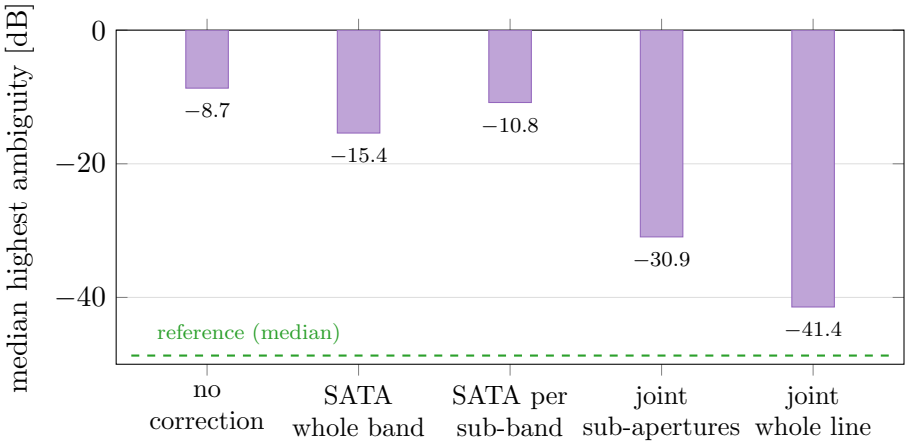


Figure 7.1: Median highest ambiguity of every method over the 680 runs of the audit. SATA per channel stays close to the uncorrected reconstruction; the joint reconstruction over the whole line gets within a few dB of the reference.

Head to head, the joint reconstruction over the whole line is better than SATA whole band in all 680 runs (median improvement 25.7 dB) and better than SATA per sub-band in all of them (median 30.5 dB). SATA per channel improves on the uncorrected reconstruction by a median of only 5.5 dB (whole band) and 0.2 dB (per sub-band), consistently with Section 4.2.1.

A distributed scene (about 11 900 scatterers, oracle −34.2 dB, measured as error-to-signal ratio) confirms the ranking: with $b_{\mathrm{xt}} \sim U(0, 100)$ m SATA per channel gives −22.5 dB to −8.7 dB and the whole-line joint −34.1 dB to −33.2 dB; with $U(0, 500)$ m, −13 dB to −6 dB and −33.6 dB to −31.4 dB.

7.2 Sweep of the final joint reconstruction in 1-D

The complete 1-D sweep (`joint_cases/sweep_joint_all.py`) runs the reference, the uncorrected reconstruction and the joint reconstruction over the whole line on 52 scenes × 20 baseline sets = 1040 runs. The medians are −47.6 dB (reference), −8.4 dB (no correction) and −40.6 dB (joint); 78 % of the runs are within 10 dB of the reference. Figure 7.2 splits the result by the number of channels and Figure 7.3 shows every run.

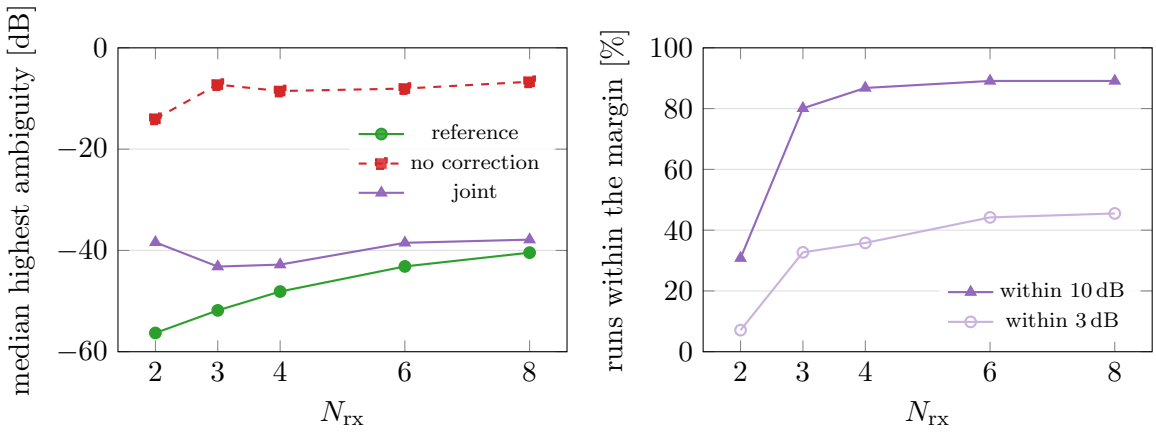


Figure 7.2: 1-D sweep by number of channels. Left: medians of the highest ambiguity. Right: fraction of runs in which the joint reconstruction is within 10 dB and 3 dB of the reference. The reference level rises with N_{rx} because $\mathrm{PRF}_{\mathrm{op}}$ falls and more ambiguity orders fall inside the band.

The 83 runs that are not resolved (above −30 dB and more than 10 dB from the reference) are almost all configurations whose baselines cannot separate the parcels: 53 have two channels and 25 the equally spaced set `s50`; only 5 have four random baselines (3 with $b_{\mathrm{xt}}{}^{\max} = 300$ m, 2 with 500 m). Figure 7.4 shows a typical resolved case on a fractal alpine relief.

7.3 Sweep of the joint reconstruction in 2-D

The 2-D sweep (`sata2d/run_sweep_joint2d_all.py`) uses 32 3-D scenes (ramps along azimuth, range and oblique; sines and cosines; realistic and fractal terrains without layover; one target on a flank, five on a diagonal or a 7 × 7 grid) with the same 20 baseline sets, 640 runs in total. The medians are −47.6 dB (reference), −13.9 dB (no correction) and −44.4 dB (joint), all in the common range band. The joint reconstruction is never worse than the uncorrected one.

With three or more channels the joint reconstruction is within a fraction of a dB of the reference in the median (Table 7.2). Of the 24 unresolved runs, 10 have two channels and 9 the equally spaced set `s50`; the others are 7 × 7 grids on a mountain, a valley or a volcano with long baselines.

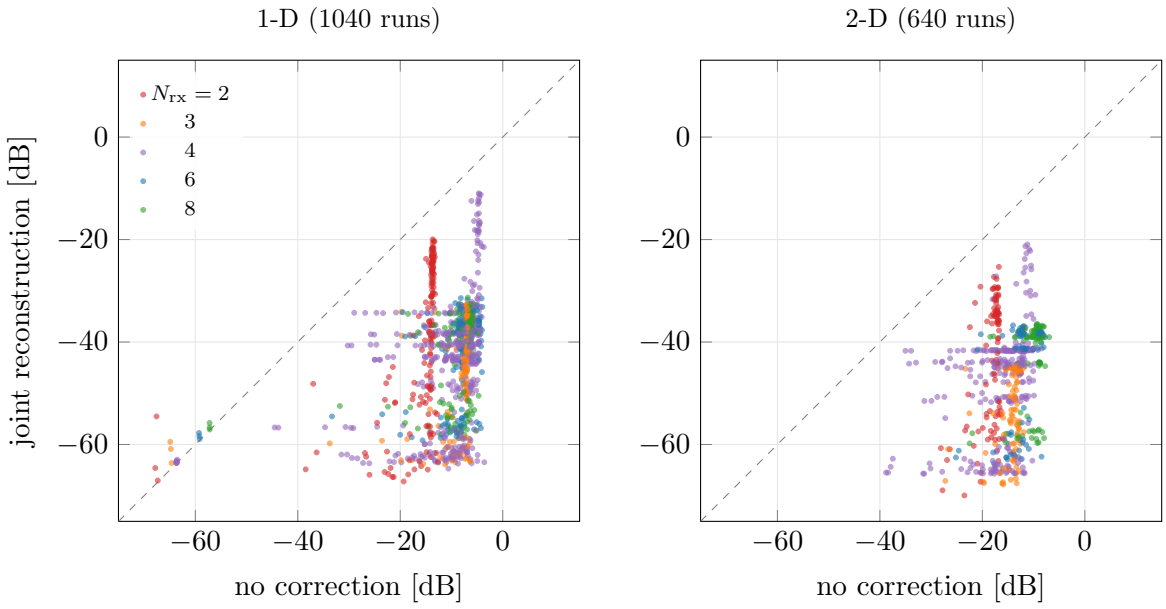


Figure 7.3: Every run of the 1-D and 2-D sweeps: highest ambiguity of the joint reconstruction against the uncorrected reconstruction, coloured by the number of channels. Points below the diagonal are improvements.

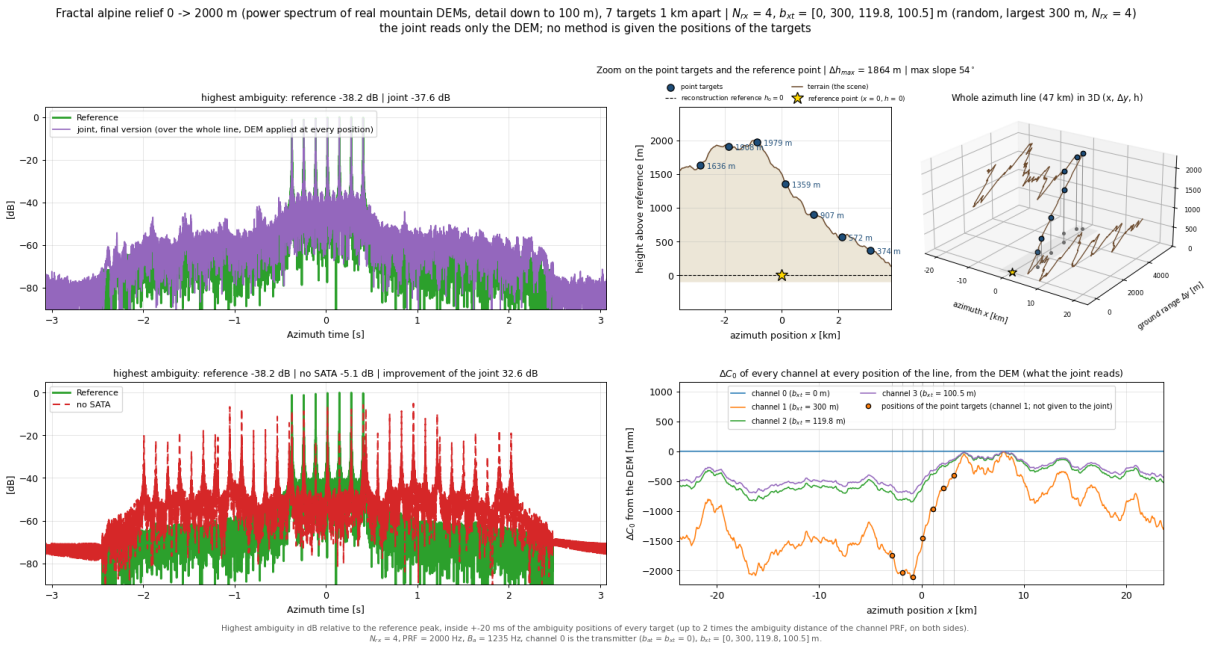


Figure 7.4: A page of the 1-D sweep: seven targets on a fractal alpine relief (0 m to 2 000 m, slopes up to 54°), $N_{\mathrm{rx}} = 4$, $b_{\mathrm{xt}} = [0, 300, 119.8, 100.5]$ m. The joint reconstruction (top left) reaches the reference level (−37.6 dB against −38.2 dB); without correction the highest ambiguity is −5.1 dB. Bottom right: the ΔC_0 of every channel along the whole line, read from the DEM.

Table 7.2: 2-D sweep by number of channels: median gap between the joint reconstruction and the reference, and fraction of runs within 3 dB of the reference.

N_{rx}	runs	reference [dB]	no corr. [dB]	joint [dB]	joint − ref. [dB]	within 3 dB [%]
2	96	−60.8	−17.4	−39.3	22.9	6
3	96	−55.5	−13.8	−53.9	1.1	65
4	256	−45.2	−15.1	−44.6	0.3	84
6	96	−41.5	−12.2	−41.4	0.2	91
8	96	−39.1	−9.5	−39.1	0.2	96

7.4 Limits

DEM errors. The phase error is $\approx 9.0 \times 10^{-5}\, b_{\mathrm{xt}}\, \delta h$ rad, so the tolerance falls with the baseline (Figure 7.5). With an rms height error of 5 m, the joint reconstruction stays at −36 dB for $b_{\mathrm{xt}} = 135$ m but rises to −20 dB for 300 m.

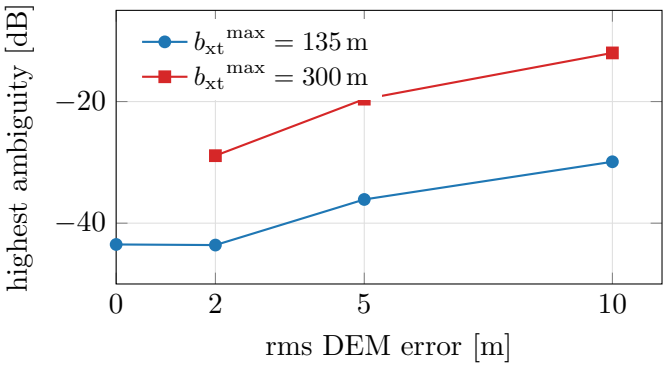


Figure 7.5: Joint reconstruction over the whole line with a random DEM error (five targets on the mountain of Figure 5.5).

To do: Confirm the scene and the error model behind Figure 7.5 (taken from the audit notes) and add the error-free point for $b_{\mathrm{xt}}^{\max} = 300$ m.

Noise. Relative to the standard reconstruction, the joint reconstruction raises the output noise by 0.3 dB to 4 dB in normal configurations, by 12 dB with equally spaced baselines and by 8 dB with two channels and 300 m. The regularisation keeps these numbers bounded where the problem is ill conditioned.

Configurations without redundancy. With two channels, or with baselines whose phases coincide at some heights, the data do not contain the information to separate the parcels. This is a property of the formation and should be considered in its design.

Not yet tested. Real antenna patterns (the model uses the simulated reference chirp), real orbits and baselines varying along the orbit, layover, distributed scenes in 2-D, noise combined with DEM errors, and real data.

8 Conclusions and outlook

8.1 Conclusions

- Cross-track baselines turn topography into a channel-dependent phase that the standard multichannel reconstruction cannot absorb, $\varphi_i \approx 2\pi b_{\mathrm{xt},i} h/(\lambda r_0 \tan\theta)$. Without correction the median highest ambiguity of the sweeps is $-8.4\,\mathrm{dB}$ in 1-D and $-13.9\,\mathrm{dB}$ in 2-D, against a reference near $-48\,\mathrm{dB}$.
- SATA computes the right phase from the DEM, but applied to each aliased channel before the reconstruction it can give only one phase to the parcels of positions $X \approx 6.2\,\mathrm{km}$ apart that share a Doppler bin. It is exact only when the terrain has the same height at all those positions; elsewhere it leaves a structural residual that is predicted from the geometry within about 1 dB, and with a DEM only it improves the median by 5.5 dB at best.
- Placing the DEM phase inside the inversion solves this. The joint reconstruction over the whole line models every channel from the focused line and the DEM and inverts it by regularised least squares. Using only the system parameters and the DEM it reaches a median of $-40.6\,\mathrm{dB}$ in 1-D (78 % of the runs within 10 dB of the reference, about 90 % for four or more channels) and $-44.4\,\mathrm{dB}$ in 2-D, where with three or more channels it is within 1 dB of the reference in the median.
- The remaining failures are formations whose baselines do not separate the folded parcels (two channels, equally spaced baselines). The regularisation leaves the ambiguity there instead of amplifying noise.
- The accuracy is limited by the DEM: the phase error grows as $b_{\mathrm{xt}}\,\delta h$, so long baselines need accurate DEMs.

8.2 Outlook

- Validation with realistic antenna patterns, orbits and baselines that vary along the orbit, and eventually with real multistatic data.
- 2-D: distributed scenes, layover and shadow, finer range resolution (envelope interpolation) and wide swaths (range-dependent kernels in blocks).
- Joint estimation of a DEM correction, using the unexplained part of the channels as the cost.
- Use of the conditioning analysis (Section 3.2) as a design rule for the cross-track baselines of a formation.
- Computational cost: block-wise or GPU implementation of the model and its adjoint for full scenes.

A Software

The code is in the repository `github.com/lucas-ayub/dlr-project`. Table A.1 maps the methods of this report to it. Generated outputs (figures, numbers) are not versioned; the numbers of Chapter 7 are extracted from the sweep outputs by `report/scripts/make_data.py`.

Table A.1: Methods and the code that implements them.

Method / task	Code
System, scene, formation	`sata1d_code/sata1d/config.py`, `geometry.py`
Signal simulation	`sata1d/signal_model.py` (`generate_reference`, `generate_channels`)
Standard reconstruction	`sata1d/reconstruction.py` (`GetCoeffNu`, `GetInversionFilters`, `reconstruct`)
SATA whole band / per sub-band	`sata1d/sata.py` (`sata_channels`), `sata1d/subband.py` (`reconstruct_subband`)
Joint in the sub-apertures	`sata1d/joint.py` (`reconstruct_joint`)
Joint over the whole line	`sata1d/joint_line.py` (`line_maps`, `reconstruct_joint_line`)
Minimal standalone joint	`sata1d_code/joint_standalone/` (package `joint1d`)
Verification and audit	`sata1d_code/verify_sata1d.py`, `joint_cases/audit_checks.py`, `joint_cases/audit_dem.py`
1-D sweep	`joint_cases/sweep_joint_all.py`
2-D model and joint	`sar_reconstruction/sata2d/` (`geometry.py`, `reconstruction.py`, `joint2d.py`)
2-D sweep	`sata2d/run_sweep_joint2d_all.py`
Reference SATA code	`runs/sata/sata_idl.txt`, `runs/sata/reference_code_SATA1D.py`

A minimal use of the final method:

```
import numpy as np
import sata1d as S

terrain  = lambda x: 1000.0 * np.exp(-x**2 / (2 * 1500.0**2))  # DEM [m]
points   = [(x, terrain(x)) for x in (-2000.0, 0.0, 2000.0)]    # (x, h)
cfg      = S.make_config(points, bxt_max=100.0, tx_rx_channel0=True)
tracks   = S.build_platform_tracks(cfg)
channels = S.generate_channels(cfg, tracks)                     # data
dC0      = S.line_maps(cfg, tracks, terrain)                    # DEM only
image    = S.reconstruct_joint_line(cfg, tracks, channels, dC0)
```

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